Name: _____ School: _____

Name: _____ Team Number: _____ Score: _____/100

The University of Texas at Austin Invitational

October 26, 2019

Water Quality B/C

Directions: You and your partner have 50 minutes to complete this test and test your salinometer at the salinometer station. You may have one sheet of notes and two non-programmable, non-graphing calculators between you and your partner. You must wear eye protection for the salinometer testing, and you may leave the event room to get eye protection if needed as long as the event has not yet started. Please be respectful of your peers at all times. If you have a question, raise your hand at your seat; do not get up. **Note: if a question in this test refers to an image, be sure to look at the image sheet, where all of the color images are.**

A. Free Response

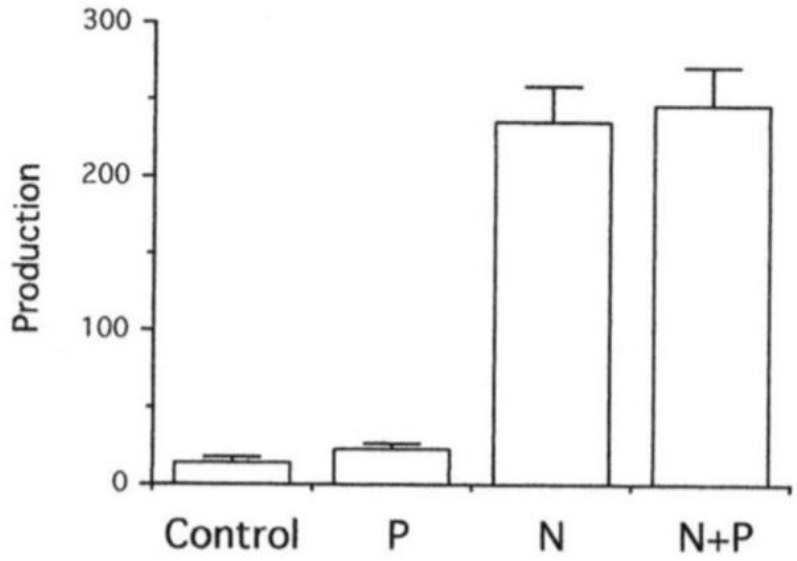
1. (3) Circle the correct answer: Ponds and lakes are (lotic/lentic) ecosystems, while rivers and streams are (lotic/lentic) ecosystems.
    a. What are the differences between lentic and lotic ecosystems?

2. (3) Refer to the image for Question 2 on the Image Sheet. Which of the pictures represents a lake that is truly nutrient rich from phosphorus and nitrogen?

3. (2) San Antonio, just an hour and a half drive south of Austin, gets most of its water from the Edwards Aquifer artesian zone. San Antonio is a city, and has a large amount of impervious cover which produces runoff, a potential pollutant to the water source. Why then is the recharge zone of the Edwards Aquifer, an area far north of San Antonio, more heavily regulated in terms of impervious cover?

4. Let's explore some biogeochemical cycles!
    a. (1) Which biogeochemical cycle does not involve the atmosphere?
    b. (4) In addition to what other element, causes hypoxia in lakes and rivers? Describe the process in detail by which this occurs.
    c. (1) (T/F) The largest reservoir of phosphorus is metamorphic rock.
    d. (1) T/F Legumes host nitrogen fixing bacteria, and thus are good crops to plant to replenish the soil
    e. (1) T/F Ammonification is the process when nitrogen containing compounds is produced from nitrogen in the soil.

5. In a theoretical experiment, four water samples are taken from a pond and their levels of productivity is measured over one week. In the control condition, no nutrients are added to the samples labeled “N”, only nitrogen is added. In the sample labeled “N+P”, both nitrogen and phosphorus is added. The graph below shows data from such an experiment.
    a. (2) Which nutrient, nitrogen or phosphorus is the limiting nutrient in the pond?
    b. (3) Why is the identity of a limiting nutrient important information for water resource managers?



6. (4) Name 3 things that are caused by water movement in estuaries
7. (3) _____ are natural polyphenol compounds that are found in plants, which protect them from predation. This bitter compounds has found its way into waterways, thereby increasing the cloudiness of water.

8. Suppose there are two different populations of African cichlids in different sections of a river. In one section of a river, resource scarcity has caused the carrying capacity to fall to 650, the same as in Section II. In Section I the current population at the start of the resource scarcity is 1000 cichlids, while in Section II the population has recently been diminished to 50 cichlids due to a disease that is now no longer afflicting the population.
    a. (5) Draw a graph below showcasing how the populations will change over time.

    b. (2) Compare the level of intraspecific competition between members of Section II's population when the population is between 50 and 100 members and when it is near 650 members.

    c. (3) African cichlids prefer aqueous environments that are basic. Why would they love any body of water with banks consisting of limestone? Describe the process that makes this environment a good fit for cichlids.

    d. (1) Suppose that some sort of pollution suddenly introduces sulfur dioxide gas and carbon dioxide gas into the river, affecting both Sections I and II. How would this affect the populations of African cichlids?

9. (3) The qualitative method of testing for phosphates in a solution depends on the fact that after a sample that contains phosphate ions is acidified with nitric acid and ammonium molybdate is added, it will produce a noticeable _____.

10. (3) How does the presence of limestone rock ($CaCO_3$) in and around a body of water affect the water's pH?

11.

a. (3) Label the classifications of the lakes, depending on the amounts of organic matter produced

   i. A:

   ii. B:

   iii. C:

b. (2) True/False. If False, correct the statement : Algal blooms and increased plant growth results in more detritus and can lead to oxygen saturation due to increased decomposition.

12. Consider the food chain shown on the Image Sheet for this question.

a. (2) Based on its position in this chain, a pike is a

   _____ order consumer

b. (3) A heron has a mass of 3 kilograms. Approximately how much biomass of diatoms is required to sustain one heron?

13. (3) Place the following steps of drinking water treatment in order:

fluoridation/chlorination, filtration, flocculation/coagulation

Answer:

_____, _____,

_____

B. Identification

1. Refer to the image shown for Question 1 in the Identification Section of the Answer Sheet

a. What is the common name of this organism? (1 points)

b. Explain the reproduction process of this organism. What does the female do to protect her eggs? (4 points)

c. What are the feeding habits of this organism? (2 points)

2. Refer to the image shown for Question 2 in the Identification Section of the Answer Sheet

a. What is the common name of this organism? (1 point)

b. Briefly describe and identify the two stages of development of this organism. Why? (2 points)

c. (T/F) This organism is an ocean dweller, and only dwells in the shallow part of the ocean (1 point)

d. Describe the defense adaptations of this organism? Is it able to regenerate body parts? Explain briefly. (3 points)

3. Refer to the image shown for Question 3 in the Identification Section of the Answer Sheet

a. What is the common name of this organism? (1 point)

b. Describe this organism's development/reproduction process? (2 points)

c. In what conditions do these organisms prefer to live in? (2 Points)

d. (T/F) This organism is extremely rare in nature and high priced in the international organmental market. (1 points)

4. **Refer to the image shown for Question 4 in the Identification Section of the Answer Sheet**

a. What is the common name of this organism? (1 point)

b. Describe the components of the skeleton of this organism. How does the fan-like shape of this organism aid feeding habits . (4 points)

c. Where do these organisms indicate unstable coral reefs? What type of conditions do these organisms prefer? (2 points)

5. **Refer to the image shown for Question 5 in the Identification Section of the Answer Sheet**

a. What is the common name of this organism? (1 points)

b. Briefly explain this organism's defining characteristics. Where do these organisms usually cluster? (2 points)

c. (T/F) It is possible that one can find a spotted and banded version of this organism because this organism’s colouring and patternings changes throughout their lives. (2 points)

6. **Refer to the image shown for Question 6 in the Identification Section of the Answer Sheet**

a. What organism is shown in the picture above? (1 point)

b. What happens when an organism comes in contact with a reef? (2 points)

c. (T/F): In high abundances these organisms provide food for a variety of herbivores. (1 point)

Water Monitoring and Analysis

1. What was your reading of salinity of the given sample in ppm? (Tiebreaker)

C. Tiebreaker

1. Fecal coliform levels are monitored because:
    a. They are disease causing agents
    b. There is a correlation between fecal coliform counts and the probability of contracting a disease from water
    c. There is a correlation between fecal coliform counts and the level of nitrites in the water
    d. Pathogens are relatively scarce in water, making them difficult to monitor directly
2. Which of the following are concerns with the presence of nitrates in surface waters? (Select all that apply)
    a. Nitrates are toxic for young humans and livestock
    b. Nitrates react reality with oxygen, lowering dissolved oxygen levels
    c. Nitrates contribute to the process of eutrophication
    d. Nitrates combine with suspended solids in water to produce sticky particles that clog water intake pipes.
3. The water quality index (WOI) is used by man to evaluate and compare waters the world over. Which of the nine tests if weighted the highest?
    a. nitrates
    b. O2
    c. pH
    d. Phosphates
4. Ammonification is the term to describe
    a. Gaseous Nitrogen to Nitrate
    b. Nitrate to Ammonia

c. Ammonia to Nitrate

d. Nitrate to Gaseous Nitrogen

5. Which of the following is not a result of cultural eutrophication of a lake?

a. Increased algal blooms

b. Increased sedimentation

c. Both A and B

d. Increased transparency

e. Decreased species diversity

f. Decreased DO concentrations